Solutions to the exercises in *Categorical Logic and Type Theory* by Bart Jacobs

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Chapter 1

Introduction to fibred category theory

1.1 Fibrations

1.1.1.

(i) Prove Proposition 1.1.4.[a]

(ii) Suppose f is Cartesian and g and h are above the same map. Show that $f \circ g = f \circ h$ implies $g = h$.

[a] [2, Proposition 1.1.4] asserts that Cartesian liftings are unique up-to-isomorphism (in a slice): if f and f' with $\operatorname{cod} f = \operatorname{cod} f'$ are both Cartesian over the same map, then there is a unique vertical isomorphism $\varphi : X \xrightarrow{\cong} X'$ with $f' \circ \varphi = f$.

We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ and $f' : X' \to Y$ are Cartesian morphisms in $\mathbb{E}$ over u. Since f' is Cartesian and f is a morphism into Y over $u = u \circ \mathbf{1}_I$, there exists a unique morphism $\varphi : X \to X'$ in $\mathbb{E}$ over I such that $f = f' \circ \varphi$. This is summarized by the following diagram.

$$
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[d] \arrow[dr, bend left, "u"] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
\quad \Longrightarrow \quad
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] \arrow[dr, dashed, "\exists!\,\varphi"] & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
$$

Similarly, there exists a unique morphism $\psi : X' \to X$ in $\mathbb{E}$ over I such that $f' = f \circ \psi$. Moreover, both $\mathbf{1}_{X'}$ and $\varphi \circ \psi$ are morphisms in $\mathbb{E}$ over I whose

composition with f' is f' itself, as suggested in the following diagram.

$$\begin{tikzcd}
X' \arrow[dr, "\psi"] \arrow[drr, bend left, "f'"] \arrow[ddr, bend right, "\mathbf{1}_{X'}"'] \arrow[dd] & & \\
& X \arrow[dr, "\varphi"] \arrow[drr, bend left, "f"] \arrow[dd] & \\
I \arrow[dr, equal] & & X' \arrow[r, "f'"] \arrow[dd] & Y \arrow[dd] \\
& I \arrow[dr, equal] & & \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, since f' is Cartesian, $\varphi \circ \psi = \mathbf{1}_{X'}$. By symmetry, $\psi \circ \varphi = \mathbf{1}_X$, and hence φ is an isomorphism. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ and $v : K \to I$ are morphisms in $\mathbb{B}$, $f : X \to Y$ is a Cartesian morphism in $\mathbb{E}$ over u, and $g, h : Z \to X$ are morphisms in $\mathbb{E}$ over v such that $f \circ g = f \circ h$. These suppositions are summarized in the following commutative diagram.

$$\begin{tikzcd}
Z \arrow[dr, bend left, "g"] \arrow[drr, bend left, "f\circ g = f\circ h"] \arrow[dr, bend right, "h"'] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Since f is Cartesian, this situation immediately implies that $g = h$. $\square$

Bibliography

[1] Benedikt Ahrens and Peter LeFanu Lumsdaine. "Displayed Categories". *Logical Methods in Computer Science* Volume 15, Issue 1 (Mar. 2019). DOI: 10.23638/LMCS-15(1:20)2019. URL: https://lmcs.episciences.org/5252.

[2] Bart Jacobs. *Categorical logic and type theory*. Vol. 141. Studies in Logic and the Foundations of Mathematics. North-Holland Publishing Co., Amsterdam, 1999, pp. xviii+760. ISBN: 0-444-50170-3.